Autodesk®
Scaleform®

Scaleform

DirectX 9 Scaleform 3D

Ben Mowery

3.09

2013 10 7

Autodesk®
GAMEWARE

Scaleform 4.3

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1

Scaleform UI
Adobe Flash Studio GPU
Scaleform Flash Studio

UI Scaleform 3D Flash Flash
3D 3D Flash UI Scaleform UI

Scaleform Flash
DirectX ShadowVolume SDK

Scaleform Scaleform
Scaleform
Scaleform

Scaleform Scaleform 4.0

DirectX
SDK ShadowVolume Scaleform “cannot create
renderer” Scaleform Player

3

3.1

Windows Scaleform ,
,Scaleform installer DirectX Scaleform C:\Program Files\Scaleform\GFx SDK 4.3

3rdParty
Scaleform libjpeg zlib

Projects
Visual Studio

App\Samples

Apps\Tutorial

Bin
Flash

FxPlayer: Flash HUD
Samples: Flash FLA

Video Demo Scaleform
Win32: Scaleform Player
AMP: AMP Flash
GFxExport: GFxExport Flash
7

Include
Scaleform

Lib
Scaleform

Resources

CLIK

Doc

2 PDF

3.2

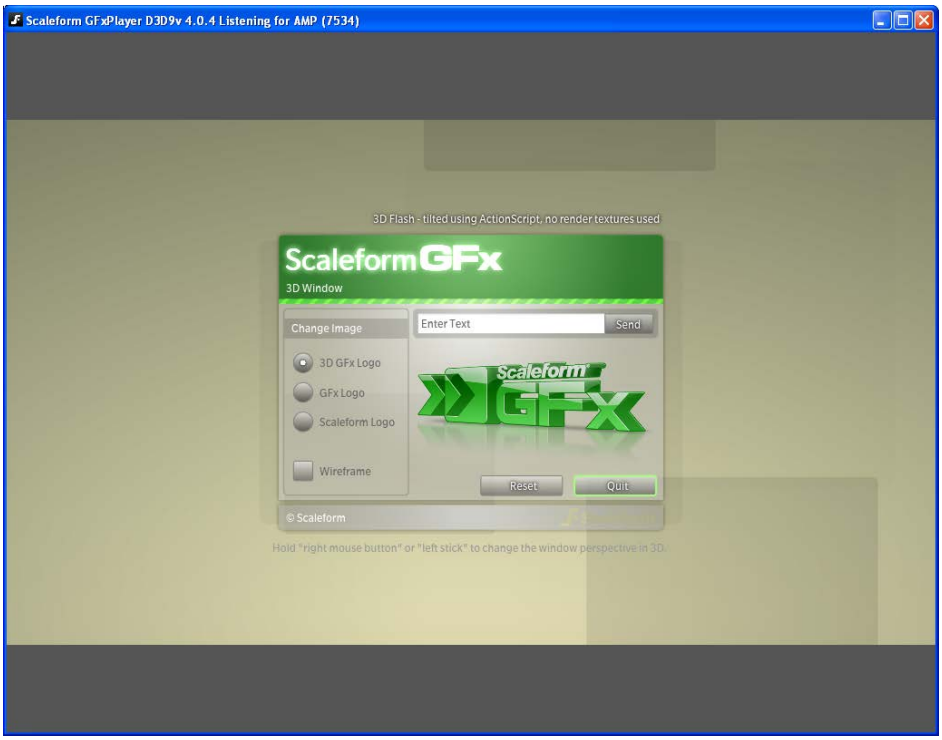
Scaleform C:\Program Files (x86)\Scaleform\GFx SDK 4.3\Projects\Win32\Msvc90\Demos\GFx 4.3 Demos.sln Demos.sln Visual Studio .sln D3D9_Debug_Static GFxPlayer

GFx SDK 4.3\Bin\Win32\Msvc90\GFxPlayer\GFxPlayer_D3D9_Debug_Static.exe 文件的更新時間，確認已成功編譯並運行該程式。

Start → All Programs → Scaleform → GFx SDK 4.3 → Demo GFx Player D3Dx SWF Scaleform Flash

3.3 Scaleform Player SWF

Start → All Programs → Scaleform → GFx SDK 4.3 → GFx Players → GFx Player D3D9 C:\Program Files\Scaleform\GFx SDK 4.3\Bin\Data\AS2\Samples\SWFToTexture 3DWindow.swf



1 Flash

F1

1. CTRL+F FPS
2. CTRL+W Scaleform Flash
   CTRL+W
3. CTRL
4. CTRL
5. CTRL+Z
6. CTRL+U
7. F2

Flash

800x600 1600x1200

Scaleform “Scaleform”
Flash Scaleform

“3D Scaleform Logo” CTRL+W
CTRL+A EdgeAA

FSAA

AA Scaleform EdgeAA

DP

EdgeAA

Scaleform Player Flash CPU CPU

Scaleform CTRL+Y

60 CPU

release debug

Scaleform debug Scaleform

3.4

Scaleform 4.3 Scaleform\GFx SDK 4.3\Apps\Tutorial SLN

Scaleform-> GFx SDK 4.3-> Tutorial Windows for Visual Studio 2005 Visual Studio 2008 “Debug”

2 ShadowVolume

3.5 *Scaleform*

Scaleform Visual Studio Tutorial
$(GFXSDK) SDK
C:\Program Files\Scaleform\GFx SDK 4.3\
$(GFXSDK) “Msvc90
Visual Studio 2008 Visual Studio 2010 Msvc90 Msvc10

Scaleform

$(GFXSDK)\Src
$(GFXSDK)\Include
Visual Studio “Additional Include Directories”

AS3 AS3 “GFx/AS3/AS3_Global.h”
AS3 Scaleform LITE

The following library directories should be added to the “Additional Library Directories” field for all build configurations:

"$(DXSDK_DIR)\Lib\x86";
"$(GFXSDK)\ 3rdParty\expat-2.1.0\lib\$(PlatformName)\Msvc90\Release";
"$(GFXSDK) \3rdParty\pcre\Lib\$(PlatformName)\Msvc90\Release”;
"$(GFXSDK)\3rdParty\zlib-1.2.7\Lib\$(PlatformName)\Msvc90\Release";
"$(GFXSDK)\3rdParty\libpng-1.5.13\Lib\$(PlatformName)\Msvc90\Release”;
"$(GFXSDK)\3rdParty\curl-7.29.0\lib\$(PlatformName)\Msvc90\Release
"$(GFXSDK)\3rdParty\jpeg-8d\Lib\$(PlatformName)\Msvc90\Debug;"
"$(GFXSDK)\Lib\$(PlatformName)\Msvc90\$(ConfigurationName)
Msvc90 Visual Studio

Scaleform
libgfx.lib
libgfx_as2.lib
libgfx_as3.lib
libgfx_air.lib
libgfxexpat.lib
libjpeg.lib
zlib.lib

libcurl.lib
libpng.lib
libgfxrender_d3d9.lib
libgfxsound_fmod.lib

.vcproj

Scaleform include Tutorial\Section3.5

3.5.1 AS3_Global.h and Obj\AS3_Obj_Global.xxx

:AS3_Global.h Obj\AS3_Obj_global.xxx
AS3_Obj_Global.xxx " "ActionScript 3 swf "
"(script) , GlobalObjectCPP, C++
Scaleform VM

AS3_Global.h ClassRegistrationTable
AS3 C++ , ,
ClassRegistrationTable, ,
AS3_Global.h

ClassRegistrationTable include
ClassRegistrationTable( ) AS3_Global.h
, (,),

, AS3 VM ,
, " " ,

4

Scaleform 3D Unreal® Engine 3, Gamebryo™, Bigworld®, Hero Engine™, Touchdown Jupiter Engine™, CryENGINE™, Trinigy Vision™ Scaleform

Scaleform DirectX DirectX ShadowVolume SDK DirectX DXUT 2D 3D

Scaleform Flash Scaleform DXUT

DXUT Win32 DirectX Scaleform SDK GFxPlayerTiny.cpp

4.1 *Flash*

Flash 3D GFx::Loader Flash GFx::MovieDef Flash GFx::Movie Render::Renderer2D Render::HAL DirectX Scaleform API /

ShadowVolume 4.1

#1:

ShadowVolume.cpp

```
#include "GFx_Kernel.h"
#include "GFx.h"
#include "GFx_Renderer_D3D9.h"
```

Scaleform GFxTutorial

Scaleform

Scaleform

```
//                  GFx::Loader
Loader           gfxLoader;

//    SWF/GFx           GFxMovieDef
Ptr<MovieDef>   pUIMovieDef;

//                    GFx::Movie
Ptr<Movie>  pUIMovie;

// Renderer
Ptr<Render::D3D9::HAL>  pRenderHAL;
Ptr<Render::Renderer2D> pRenderer;
MovieDisplayHandle      hMovieDisplay;
```

#2 GFx::System

Scaleform GFx::System Scaleform WinMain

```
//                  GFx::System
GFx::System          gfxInit;
```

GFx::System Scaleform Scaleform

WinMain GFx::System Scaleform

Tutorial

GFx::System

GFx::System GFxTutorial

GFx::System

GFx::System

GFx::System::Init() GFx::System::Destroy()

#3:

Scaleform WinMain InitApp()

InitApp()

```
gfx = new GFxTutorial();
assert(gfx != NULL);
if(!gfx->InitGFx())
      assert(0);
```

GFxTutorial GFx::Loader GFx::Loader
SWF/GFx SWF/GFx
SWF/GFx GFx::Loader
GFx::Log

GFxTutorial::InitGFx() GFx::Loader GFx::Loader
SetLog Scaleform
GFxPlayerLog
GFx::Log

```
//          —Scaleform
//
gfxLoader->SetLog(Ptr<Log>(*new PlayerLog()));
```

GFx::Loader GFx::FileOpener
GFx::FileOpener

```
//
Ptr<FileOpener> pfileOpener = *new FileOpener;
gfxLoader->SetFileOpener(pfileOpener);
```

Render::HAL Scaleform D3D9
D3D Scaleform
InitHAL Render HAL IDirect3Device9
Scaleform DX9 UI

```
pRenderHAL = *new Render::D3D9::HAL();
if (!(pRenderer = *new Render::Renderer2D(pRenderHAL.GetPtr())))
     return false;
```

Scaleform Render::HAL Render::D3D9::HAL
Scaleform

#4 Flash

GFx::Loader GFx::MovieDef GFx::MovieDef

```
//
pUIMovieDef = *gfxLoader.CreateMovie(UIMOVIE_FILENAME,
                                     Loader::LoadKeepBindData |
                                     Loader::LoadWaitFrame1, 0);
```

LoadKeepBindData D3D

LoadWaitFrame1 CreateMovie GFx::ThreadTaskManager

Memory System Overview

#5

GFx::MovieDef GFx::Movie GFx::Movie

```
pUIMovie = *pUIMovieDef->CreateInstance(true, 0, NULL);
assert(pUIMovie.getPtr() != NULL);
```

CreateInstance false Flash Memory System Overview

Advance() false CreateInstance

```
//
```

```
pUIMovie->Advance(0.0f, 0, true);

//
MovieLastTime = timeGetTime();
```

Advance

3D

```
pUIMovie->SetBackgroundAlpha(0.0f);
```

Flash 3D

#6

Scaleform Render::D3D9::HAL DirectX
D3D Scaleform Render::D3D9::HAL::InitHAL 。
/ D3D InitHAL

ShadowVolume OnResetDevice DXUT
GFxTutorial OnResetDevice

```
pRenderHAL->InitHAL(
    Render::D3D9::HALInitParams(pd3dDevice, presentParams,
                                Render::D3D9::HALConfig_NoSceneCalls));
```

InitHAL() D3D Scaleform HAL_NoSceneCalls
Scaleform DirectX BeginScene() EndScene()
ShadowVolume OnFrameRender

#7

D3D D3D
ShadowVolume OnLostDevice
Render::HAL GFxTutorial OnLostDevice D3D

```
pRenderHAL->ShutdownHAL();
```

SetViewport PC

Scaleform

OpenGL

Scaleform Flash 4.2

#10 DirectX

ShadowVolume OnFrameRender() D3D
BeginScene() EndScene() EndScene()
GFxTutorial::AdvanceAndRender()

```
void AdvanceAndRender(void)
{
      DWORD mtime = timeGetTime();
      float deltaTime = ((float)(mtime - MovieLastTime)) / 1000.0f;
      MovieLastTime = mtime;

      pUIMovie->Advance(deltaTime, 0);
      pRenderer->BeginFrame();

      if (hMovieDisplay.NextCapture(pRenderer->GetContextNotify()))
      {
          pRenderer->Display(hMovieDisplay);
      }

      pRenderer->EndFrame();
}
```

Advance() deltaTime
GFx::Movie::Advance

#11

Render::Renderer2D::Display DirectX D3D
D3D
Render::Renderer2D::Display D3D

Scaleform

DX9

DX9 GFxTutorial::AdvanceAndRender()

```
//     GFx     DirectX
g_pStateBlock->Capture();

//
gfx->AdvanceAndRender();

//     DirectX
g_pStateBlock->Apply();
```

#12 UI

DXUT 4.1

ShadowVolume.cpp DXUT

```
// Disable default UI
...
```

DirectX Flash

SetViewAlignment SetViewScaleMode SM_NoScale

Scaleform

d3d9guide.fla

ActionScript C++ 。SetViewScaleMode

SetViewAlignment ActionScript Stage (State.scaleMode, Stage.align)。

4.3

ShadowVolume Scaleform Flash Flash

GFx::Movie::HandleEvent GFx::Event

GFx::Movie

4.3.1

ShadowVolume MsgProc Win32 GFxTutorial::ProcessEvent

Scaleform WM_MOUSEMOVE,

WM_LBUTTONDOWN WM_LBUTTONUP

```
void ProcessEvent(HWND hWnd, unsigned uMsg, WPARAM wParam, LPARAM lParam,
                  bool *pbNoFurtherProcessing)
{
      int mx = LOWORD(lParam), my = HIWORD(lParam);
      if(pUIMovie)
      {
            if(uMsg == WM_MOUSEMOVE)
            {
                  MouseEvent mevent(GFx::Event::MouseMove, 0, mx, my);
                  pUIMovie->HandleEvent(mevent);
            }
            else if(pMovieButton && uMsg == WM_LBUTTONDOWN)
            {
                  ::SetCapture(hWnd);
                  MouseEvent mevent(GFx::Event::MouseDown, 0, mx, my);
                  pUIMovie->HandleEvent(mevent);
            }
            else if(pMovieButton && uMsg == WM_LBUTTONUP)
            {
                  ::ReleaseCapture();
                  MouseEvent mevent(GFx::Event::MouseUp, 0, mx, my);
                  pUIMovie->HandleEvent(mevent);
```

```
                }
            }
        }
```

Scaleform

#1:

```
pMovie->SetViewport(screen_width, screen_height, 0, 0, screen_width,
                                                    screen_height, 0);
```

(0, 0)

Scaleform

#2

```
pMovie->SetViewport(screen_width, screen_height, 0, 0, screen_width / 4,
                                                screen_height / 4, 0);
```

HandleEvent

Scaleform

#3

```
movie_width = screen_width / 6;
movie_height = screen_height / 6;
pMovie->SetViewport(screen_width, screen_height,
                            screen_width / 2 – movie_width / 2,
                          screen_height / 2 – movie_height / 2,
                    movie_width, movie_height);
```

(0,0) (screen_width / 2 – movie_width / 2, screen_height / 2 – movie_height / 2)

Flash GFx::Movie::SetViewAlignment GFx::Movie::SetViewport SetViewAlignment SetViewScaleMode Scaleform

4.3.2

GFx::Movie:HandleEvent GFx::KeyEvent GFx::CharEvent:

```
KeyEvent(EventType eventType = None,
                  Key::Code code = Key::None,
                  UByte asciiCode = 0,
                  UInt32 wcharCode = 0,
                  UInt8 keyboardIndex = 0)

CharEvent(UInt32 wcharCode, UInt8 keyboardIndex = 0)
```

GFx::KeyEvent GFx::CharEvent ASCII Windows GFx::CharEvent WM_CHAR GFx::KeyEvents WM_SYSKEYDOWN, WM_SYSKEYUP, WM_KEYDOWN WM_KEYUP

- 'c' SHIFT : GFx::KeyEvent WM_KEYDOWN
  - 'c'
  -
  - SHIFT
- GFx::CharEvent WM_CHAR " " ASCII 'C' Scaleform
- 'c' GFx::KeyEvent WM_KEYUP GFx::CharEvent
- F5 GFx::KeyEvent F5 ASCII GFx::CharEvent

GFx::KeyEvent GFx_Event.h Flash GFxPlayerTiny.cpp Scaleform Player Windows Flash ProcessKeyEvent 3D

```
void ProcessKeyEvent(Movie *pMovie, unsigned uMsg, WPARAM wParam, LPARAM
lParam)
```

Windows WndProc WM_CHAR, WM_SYSKEYDOWN, WM_SYSKEYUP, WM_KEYDOWN WM_KEYUP Scaleform pMovie

GFx::KeyEvent GFx::CharEvent GFx::CharEvent ASCII Unicode IME . IME IME GFx::KeyEvent Page Up Page Down

4.3 3D “Settings” DX9 C++ d3d9guide.fla

Flash UI UI Scaleform DXUT DXUT Scaleform Scaleform DXUT Scaleform Flash UI

“Change Mesh” “Change Mesh” Flash

4.3.3

3D

UI 3D GFx::Movie::HitTest 解决 Flash 中的 在处理完一个鼠标事件后，改变 GFxTutorial::ProcessEvent 以调用 HitTest， UI DXUT

```
bool processedMouseEvent = false;
if(uMsg == WM_MOUSEMOVE)
{
```

```
            MouseEvent mevent(GFx::Event::MouseMove, 0, (float)mx, (float)my);
            pUIMovie->HandleEvent(mevent);
            processedMouseEvent = true;
        }
        else if(uMsg == WM_LBUTTONDOWN)
        {
            ::SetCapture(hWnd);
            MouseEvent mevent(GFx::Event::MouseDown, 0, (float)mx, (float)my);
            pUIMovie->HandleEvent(mevent);
            processedMouseEvent = true;
        }
        else if(uMsg == WM_LBUTTONUP)
        {
            ::ReleaseCapture();
            MouseEvent mevent(GFx::Event::MouseUp, 0, (float)mx, (float)my);
            pUIMovie->HandleEvent(mevent);
            processedMouseEvent = true;
        }

        if(processedMouseEvent && pUIMovie->HitTest((float)mx, (float)my,
            Movie::HitTest_Shapes))
            *pbNoFurtherProcessing = true;
```

4.3.4

"Change Mesh"

4.3.2 W,S,A,D Q 3D Scaleform 3D

6.1.2 C++

4.3.5

, , Windows API Scaleform Windows 7, , ,

, **WINVER=0x601** Windows API Windows 7 ( ) ShadowVolume.cpp , `#define ENABLE_MULTITOUCH` ,

, #include <windows.h> ,
RegisterTouchWindow , RegisterTouchWindow MsgProc
WM_TOUCH

```
if(uMsg == WM_TOUCH)
{
      ProcessTouchEvent(pUIMovie, uMsg, wParam, lParam);
}
```

ProcessTouchEvent WM_TOUCH Windows
GFx::TouchEvent HandleEvent GFx::TouchEvent
ID). x/y 0..1)

```
TouchEvent( EventType evtType,
            unsigned id,
            float _x,
            float _y,
            float wcontact = 0,
            float hcontact = 0,
            bool primary = true,
            float pressure = 1.0f)
```

SWF , ,GFx 0
MultitouchInterface, , ,
GFx::GestureEvent(, ,
Scaleform

```
class FxPlayerMultitouchInterface : public MultitouchInterface
{
public:
    //
    virtual unsigned GetMaxTouchPoints() const { return 2; }

    //
    virtual UInt32   GetSupportedGesturesMask() const { return 0; }

    //
    virtual bool     SetMultitouchInputMode(MultitouchInputMode) { return
true; }
};
```

SetMultitouchInterface ,

5

5.1

Scaleform

Flash

TTF

Windows Linux Scaleform

http://gameware.autodesk.com/scaleform/developer/?action=doc

5000

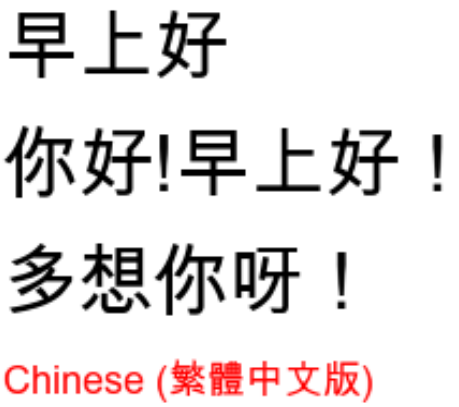
Scaleform

Scaleform

Scaleform TTF

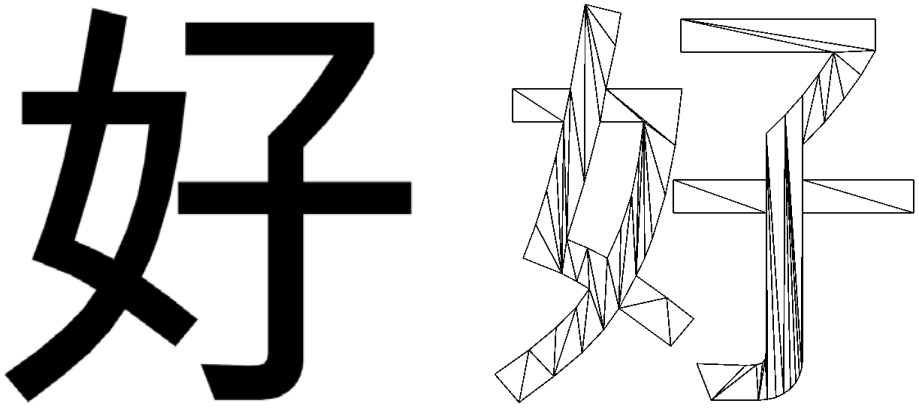
Scaleform

Bin\Data\AS2\Samples\FontConfig sample.swf Scaleform Player D3D9



3a **4b**

CTRL+W



4c **4d**

4c 4d Scaleform

Flash

gfx_2.1_texteffects_sample.zip

Isotropic shadow with offset
Text Shadow
文本陰影
Anisotropic glow, BlurX=2, BlurY=7
Text Glow 1
文本煥發 1
Isotropic glow, BlurX=BlurY=4, Knockout
TEXT GLOW 2
文本煥發 2
Rotating glow
Text Glow 3
文本煥發 3
Changing blurriness
Text Glow 4
文本煥發 4

5

6 C++ Flash

Flash

Flash

Scaleform C++

Scaleform

C++

6.1 C++

Scaleform C++ FSCommand
ExternalInterface FSCommand ExternalInterface GFx::Loader C++
FSCommand fscommand
ExternalInterface flash.external.ExternalInterface.call
GFx::Value 6.1.2

FSCommands ExternalInterface
fscommands gfxexport SWF
fscommands –fstree –fslist –fsparams ExternalInterface
fscommands

6.1.1 FSCommand

fscommand

```
fscommand("setMode", "2");
```

fscommand ExternalInterface

Scaleform FSCommand GFx::FSCommandHandler
FSCommand GFx::Loader GFx::Movie
GFx::Movie fscommand GFxPlayerTiny
(“FxPlayerFSCommandHandler”) ShadowVolume
6.1

GFx::FSCommandHandler

```
class OurFSCommandHandler : public FSCommandHandler
{
    public:
    virtual void Callback(Movie* pmovie,
                          const char* pcommand, const char* parg)
    {
        GFxPrintf("FSCommand: %s, Args: %s", pcommand, parg);
    }
};
```

Callback fscommand fscommand

GFxTutorial::InitGFx() GFx::Loader

```
//      FSCommand
Ptr<FSCommandHandler> pcommandHandler = *new OurFSCommandHandler;
gfxLoader->SetFSCommandHandler(pcommandHandler);
```

GFx::Loader GFx::Movie SetFSCommandHandler

fscommand ShadowVolume
"Toggle Fullscreen"

```
FSCommand: ToggleFullscreen, Args:
```

Open Flash/HUDMgr.as in Flash Studio. This class is referenced from d3d9guideAS3.fla. The association between this external class and the Flash content is made by setting the ActionScript class for the "hudMgr" Symbol in d3d9guideAS3's Symbol Library.

Compare the events printed to the screen with the fscommand() calls made from the HUDMgr class in the following function:

```
function toggleFullscreen(ev:MouseEvent) {
                    fscommand("ToggleFullscreen");
}
```

As an exercise, change fscommand("ToggleFullscreen") to fscommand("ToggleFullscreen", String(hud.visible)) to print the value of the hud.visible value to C++. Export the flash movie (CTRL+ALT+Shift+S) and replace d3d9guideAS3.swf.

FSCommand ShadowVolume
fscommand

```
if(strcmp(pcommand, "ToggleFullscreen") == 0)
      doToggleFullscreen = true;
```

DXUT OnFrameMove OnFrameMove

```
if(doToggleFullscreen)
{
      doToggleFullscreen = false;
      DXUTToggleFullScreen();
}
```

Advance
Invoke 。

6.1.2 ExternalInterface

Flash ExternalInterface fscommand

ExternalInterface fscommand

```
class OurExternalInterfaceHandler : public ExternalInterface
{
  public:
      virtual void Callback(Movie* pmovieView,
                            const char* methodName,
                            const Value* args,
                            unsigned argCount)
      {
            Printf("ExternalInterface: %s, %d args: ",
                                        methodName, argCount);
            for(unsigned i = 0; i < argCount; i++)
            {
                  switch(args[i].GetType())
                  {
                    case Value::VT_Null:
                        GFxPrintf("NULL");
                        break;
```

```
                        case Value::VT_Boolean:
                            GFxPrintf("%s", args[i].GetBool()? "true" : "false");
                            break;
                        case Value::VT_Int:
                            GFxPrintf("%s", args[i].GetInt());
                            break;
                        case Value::VT_Number:
                            GFxPrintf("%3.3f", args[i].GetNumber());
                            break;
                        case Value::VT_String:
                            GFxPrintf("%s", args[i].GetString());
                            break;
                        default:;
                            GFxPrintf("unknown");
                            break;
                    }
                    GFxPrintf("%s", (i == argCount - 1) ? "" : ", ");
                }
                GFxPrintf("\n");
            }
        };
```

GFx::Loader

```
        Ptr<ExternalInterface> pEIHandler = *new OurExternalInterfaceHandler;
        gfxLoader.SetExternalInterface(pEIHandler);
```

Open d3d9HUD.as used by d3d9guideAS3.fla and look at the ActionScript for the following functions:

```
        function onSubmit(path:String) {
            ExternalInterface.call("MeshPath", path);
            CloseMeshPath();
        }

        function onFocusIn(ev: FocusHandlerEvent) {
            ExternalInterface.call("MeshPathFocus", true);
        }

        function onFocusOut(ev: FocusHandlerEvent) {
            ExternalInterface.call("MeshPathFocus", false);
        }
```

MeshPathFocus

(true false) “MeshPathFocus”

```
Callback! MeshPathFocus, nargs = 1
  arg(0) = true

Callback! MeshPathFocus, nargs = 1
   arg(0) = false
```

GFxTutorial

```
if(strcmp(methodName, "MeshPathFocus") == 0 && argCount == 1 &&
      args[0].GetType() == Value::VT_Boolean)   {
            if (args[0].GetType() == Value::VT_Boolean)
                  gfx->SetTextboxFocus(args[0].GetBool());
      }
```

GFxTutorial::ProcessEvent
3D

```
      if(uMsg == WM_SYSKEYDOWN || uMsg == WM_SYSKEYUP ||
         uMsg == WM_KEYDOWN    || uMsg == WM_KEYUP    ||
         uMsg == WM_CHAR)
      {
            if(textboxHasFocus || wParam == 32 || wParam == 9)
            {
                  ProcessKeyEvent(pUIMovie, uMsg, wParam, lParam);
                  *pbNoFurtherProcessing = true;
            }
      }
```

(ASCII 32) tab(ASCII 9) “Toggle UI” “Settings”

“Change Mesh”

OurExternalInterfaceHandler::Callback

```
      static bool doChangeMesh = false;
      static wchar_t changeMeshFilename[MAX_PATH] = L"";

      ...

      if(strcmp(methodName, "MeshPath") == 0 && argCount == 1)
      {
```

```
        doChangeMesh = true;
        const char *filename = args[0].GetString();
        MultiByteToWideChar(CP_ACP, MB_PRECOMPOSED, filename, -1,
              changeMeshFilename, _countof(changeMeshFilename));
  }
```

DXUT OnFrameMove DXUT

6.2 C++

6.1 C++ Scaleform

C++ Scaleform C++

6.2.1

Scaleform GetVariable SetVariable 6.2

HUD (fxplayer.swf) Scaleform Player F5

```
void GFxTutorial::ProcessEvent(HWND hWnd, unsigned uMsg, WPARAM wParam,
                          LPARAM lParam, bool *pbNoFurtherProcessing)
{
      int mx = LOWORD(lParam), my = HIWORD(lParam);

      if(pHUDMovie && uMsg == WM_KEYDOWN)
      {
            if(wParam == VK_F5)
            {
                  int counter = (int)pHUDMovie-
                                          >GetVariableDouble("_root.counter");
                  counter++;
                  pHUDMovie->SetVariable("_root.counter",
                                            Value((double)counter));
                  char str[256];
                  sprintf_s(str, "testing! counter = %d", counter);
                  pHUDMovie->SetVariable("_root.MessageText.text", str);
            }
      }
      ...
```

GetVariableDouble _root.counter GetVariableDouble
SetVariable _root.counter
GFx::Movie GetVariable SetVariable

fxplayer Flash MessageText
HUDText _root.counter SetVariable

Flash TextField.text GFx::Movie::Invoke
6.2.2
Scaleform

8 SetVariable HUD

GFx::Value

SetVariable GFx::Movie::SetVarType

_root.mytextfield 3 SetVariable("_root.mytextfield.text", "testing",
SV_Normal) 1 SV_Sticky
3 _root.mytextfield.text C++

SetVariableArray Flash
GFxTutorial::InitGFx() _root.SceneData

```
//
Value sceneData[3];
sceneData[0].SetString("Scene with shadow");
sceneData[1].SetString("Show shadow volume");
sceneData[2].SetString("Shadow volume complexity");
pUIMovie->SetVariableArraySize("_root.SceneData", 3);
pUIMovie->SetVariableArray(Movie::SA_Value,
      "_root.SceneData", 0, sceneData, 3);
```

6.1
ShadowVolume

C++
C++

6.2.2

GFx::Movie::Invoke
UI

GFx::Movie::Invoke “Change Mesh” F6 F7
SetVariable
SetVariable Invoke ActionScript

GFx::Movie::Invoke WM_CHAR openMeshPath

```
...
else if (wParam == VK_F6)
{
      bool retval = pHUDMovie->Invoke("root.ui.hud.OpenMeshPath", "");
      GFxPrintf("root.ui.hud.OpenMeshPath returns '%d'\n", (int) retval);
}
else if (wParam == VK_F7)
{
      const char *retval = pHUDMovie->Invoke(
```

```
                                        "root.ui.hud.CloseMeshPath", "");
        GFxPrintf("root.ui.hud.CloseMeshPath returns '%d'\n", (int) retval);
}
```

```
            loader.load( new URLRequest( url ) );
            addChild( loader )
        }
```

loadMovie

，root.inventoryWindow.widget1.counter）。

Flash root root

level 6，root.counter level6.counter

loadMovie root.myMovie root.counter

：lockroot = true。Lockroot ActionScript

root root root ActionScript

。 Adobe Flash

GFx::Movie

Container.fla

loadMovie loadClip

Loader Container.fla

ActionScript ExternalInterface C++ 。

```
function loadMovie(url:String):void {
        var loader = new Loader();
        loader.load( new URLRequest( url ) );
        loader.contentLoaderInfo.addEventListener( Event.OPEN, handleLoadOpen,
                                                   false, 0, true );
        loader.contentLoaderInfo.addEventListener( Event.INIT, handleLoadInit,
                                                   false, 0, true );
        loader.contentLoaderInfo.addEventListener( Event.PROGRESS,
                                        handleLoadProgress, false, 0, true );
        loader.contentLoaderInfo.addEventListener( Event.COMPLETE,
                                        handleLoadComplete, false, 0, true );
        addChild( loader )
    }

    function handleLoadOpen( e:Event ):void {
```

```
    trace("Event.OPEN - Load Started!");
}
function handleLoadInit( e:Event ):void {
    trace("Event.INIT - Loaded Content Ready!");
}
function handleLoadProgress( e:Event ):void {
    trace("Event.PROGRESS - Load Progress!");
}
function handleLoadComplete( e:Event ):void {
    trace("Event.COMPLETE - Load Complete!");
}
```

7

3D , , , " " , Flash Studio SWF , SWF ,

, UI , UI

1. , cell.x
, (cellwall.jpg)
UI , ,
cellwallRT.jpg , MeshMaterialList

```
Material {
1.000000;1.000000;1.000000;1.000000;;
40.000000;
```

```
1.000000;1.000000;1.000000;;
0.000000;0.000000;0.000000;;
TextureFilename {
"cellwallRT.jpg";
}
```

2. ,cell.x DXUT ,DXUT
, CreateTexture
; , (Render Texture) ,
DXSDK ,
, DXUT ,

```
pd3dDevice->CreateTexture(rtw,rth,0,
D3DUSAGE_RENDERTARGET|D3DUSAGE_AUTOGENMIPMAP, D3DFMT_A8R8G8B8,
            D3DPOOL_DEFAULT, &g_pRenderTex, 0);

g_Background[0].m_pTextures[BACK_WALL] = g_pRenderTex;
ptex->Release();
```

3. AdvanceAndRender GFx ,

   a. ,
   ```
   pd3dDevice->GetRenderTarget(0, &poldSurface);
   pd3dDevice->GetDepthStencilSurface(&poldDepthSurface);
   ```
   b. , ,
   ```
   ptex->GetSurfaceLevel(0, &psurface);
   HRESULThr = pd3dDevice->SetRenderTarget(0, psurface );
   ```
   c. , Display, a

8 OpenGL

OpenGL/GLUT (Tutorial\OpenGL), Nvidia (Tessellation)
(HTTP://developer.download.nvidia.com/SDK/10/opengl/samples.html#tessellation)
, D3D9 tessellation.cpp tessellation_original.cpp
, NVIDIA (Init() ShutDown
AdvanceAndDisplay()) GFx , GFx GFxPlayerImpl
(GFxPlayerGL.cpp)

9 GFxExport

SWF SWF
SWF SWF
SWF

GFxExport SWF
DDS DXT

GFxExport

Scaleform GFxExport
d3d9guide.swf fxplayer.swf Scaleform

```
gfxexport -i DDS -c d3d9guideAS3.swf
gfxexport -i DDS -c fxplayer.swf
```

gfxexport.exe C:\Program Files\Scaleform\$(GFXSDK)
convert.bat 7

-i DDS DDS DirectX DXT
DXT

-c Scaleform
DXT

-share_images

ShadowVolume 8 Scaleform GFx::Loader::CreateMovie

10

Scaleform

- Flash Scaleform Player SWF SDK Gamebryo 3
- FAQs
- Scale9 Scale9Grid Overview
- Scaleform SWF Flash GFx::FileOpener
- GFx::ImageCreator
- GFx::Translator